The Dirichlet Pólya Problem for Spherical Shells in General Dimension:
Exact Dimension Lifting and the Shifted-Tail Bottleneck

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Abstract

For a spherical shell in dimension $d \geq 3$, the separated radial problem has Bessel orders $\ell + (d-2)/2$ and spherical-harmonic multiplicities $\kappa_{d,\ell}$. We give an exact branching decomposition of the strict shell-action proxy into positive integer- and half-integer shifted planar tails. We also prove a signed primitive comparison and retain its full positive remainder. This yields the exact identity

$$W_d - P_d^{<} = \mathcal{B}_d(A) + \sum_{m \geq 0} \binom{m+d-3}{d-3} D_A\left(m + \frac{d-2}{2}\right).$$

Consequently, one dimension-free positive shifted-tail inequality implies the sharp Dirichlet Pólya inequality for every spherical shell and every $d \geq 3$. The dimension lift is proved here; the inner shifted-tail inequality remains the central open lemma. We prove the outer and terminal tail regions and reduce the first interface-cell milestone to a bounded nine-shift residual family. No assertion in this manuscript treats the remaining shifted-tail lemma as proved.

1 Statement and status

For $0 < \rho < 1$ let

$$A_{\rho,1}^{(d)} = \{x \in \mathbb{R}^d : \rho < |x| < 1\}, \qquad d \geq 3,$$

and use the strict counting convention

$$N_D(A_{\rho,1}^{(d)}, K^2) := \#\{j : \lambda_j^D(A_{\rho,1}^{(d)}) < K^2\}.$$

Write

$$\omega_d = \frac{\pi^{d/2}}{\Gamma(d/2+1)}, \qquad L_d = \frac{\omega_d}{(2\pi)^d}.$$

The desired estimate is

$$N_D(A_{\rho,1}^{(d)}, K^2) \leq L_d |A_{\rho,1}^{(d)}| K^d = \frac{1-\rho^d}{2^d \Gamma(d/2+1)^2} K^d. \tag{1}$$

For $a > 0$, define the zero-extended ball action

$$G_a(z) = \begin{cases} \dfrac{1}{\pi}\left(\sqrt{a^2 - z^2} - z \arccos \dfrac{z}{a}\right), & 0 \leq z \leq a, \\ 0, & z > a, \end{cases} \tag{2}$$

put $\mu = \rho K$, and set

$$A(z) = A_{\rho,K}(z) := G_K(z) - G_\mu(z). \tag{3}$$

For $x > 0$, let

$$[x]_< := \#\{n \in \mathbb{N} : n < x\} = \max\{0, \lceil x \rceil - 1\}.$$

In particular, $[n]_< = n - 1$ at a positive integral wall.

The only unproved statement on the primary path is the following.

Conjecture 1.1 (Positive shifted shell tails)**.** *For $0 < \rho < 1$, $K > 0$, and*

$$r \in \tfrac{1}{2}\mathbb{N},$$

one has

$$T_A(r) := [A(r) + \tfrac{1}{4}]_< + 2\sum_{j\geq 1}[A(r+j) + \tfrac{1}{4}]_< \leq 2\int_r^K A(z)\,\mathrm{d}z. \tag{4}$$

The zero extension makes the sum finite and makes the assertion trivial when $r \geq K$.

Sections 2–5 prove that Conjecture 1.1 implies (1) in every dimension $d \geq 3$. Section 6 records the proved tail regions and a sharper reduction of the first interface-cell problem.

2 General-dimensional spectral proxy

Put

$$a_d = \frac{d-2}{2}, \qquad \nu_{\ell,d} = \ell + a_d.$$

The degree-ℓ spherical harmonics on $\mathbb{S}^{d-1}$ have multiplicity

$$\kappa_{d,\ell} = \binom{\ell+d-1}{d-1} - \binom{\ell+d-3}{d-1} = \frac{(2\ell+d-2)(\ell+d-3)!}{\ell!(d-2)!}. \tag{5}$$

Here and below a binomial coefficient with lower index larger than its nonnegative upper index is zero.

Under the unitary radial substitution

$$u(r) = r^{(d-1)/2}R(r),$$

the degree-ℓ block becomes

$$L_{\ell,d} = -\frac{\mathrm{d}^2}{\mathrm{d}r^2} + \frac{\nu_{\ell,d}^2 - \frac{1}{4}}{r^2} \quad \text{on } (\rho, 1), \qquad u(\rho) = u(1) = 0. \tag{6}$$

Its positive eigenfrequencies are the zeros of

$$D_{\nu,\rho}(k) = J_\nu(\rho k)Y_\nu(k) - Y_\nu(\rho k)J_\nu(k), \qquad \nu = \nu_{\ell,d}.$$

With the standard continuous Bessel phase, let

$$\gamma_{\rho,K}(\nu) = \frac{\theta_\nu(K) - \theta_\nu(\rho K)}{\pi}.$$

The determinant factorization and phase monotonicity give the exact strict count

$$N_D(A^{(d)}_{\rho,1}, K^2) = \sum_{\ell\geq 0} \kappa_{d,\ell}[\gamma_{\rho,K}(\nu_{\ell,d})]_<. \tag{7}$$

The real-order phase-difference enclosure of [2, 3] gives, for $0 \leq \nu \leq K$,

$$\gamma_{\rho,K}(\nu) < A_{\rho,K}(\nu) + \frac{1}{4}. \tag{8}$$

For $\nu \geq K$, both the exact count and the action proxy vanish. Since $x < y$ implies $[x]_< \leq [y]_<$, including when y is integral, (7)–(8) imply the wall-safe proxy bound

$$N_D(A^{(d)}_{\rho,1}, K^2) \leq P^<_d(\rho, K), \qquad P^<_d := \sum_{\ell\geq 0} \kappa_{d,\ell} q_\ell, \tag{9}$$

where

$$q_\ell = [A(\ell + a_d) + \frac{1}{4}]_<. \tag{10}$$

The argument above is dimension-independent after the substitutions $\nu = \ell + a_d$ and (5). In particular, the published phase enclosure is already a real-order result: no interpolation is introduced when d changes parity.

3 Exact harmonic branching

Let $p = d - 2$, so $a_d = p/2$, and define

$$r_m = m + \frac{p}{2}, \qquad c_{d,m} = \binom{m+p-1}{p-1} = \binom{m+d-3}{d-3}. \tag{11}$$

Proposition 3.1 (Exact shifted-tail decomposition)**.** *For every finitely supported sequence $(q_\ell)_{\ell\geq 0}$,*

$$\sum_{\ell\geq 0} \kappa_{d,\ell} q_\ell = \sum_{m\geq 0} c_{d,m} \left(q_m + 2\sum_{j\geq 1} q_{m+j} \right). \tag{12}$$

Consequently, for (10),

$$\boxed{P^<_d = \sum_{m\geq 0} c_{d,m} T_A(r_m).} \tag{13}$$

This identity is exact at every strict action wall.

Proof. The generating functions are

$$\sum_{\ell\geq 0} \kappa_{d,\ell} s^\ell = \frac{1+s}{(1-s)^{d-1}}, \qquad \sum_{m\geq 0} c_{d,m} s^m = \frac{1}{(1-s)^{d-2}},$$

and

$$1 + 2\sum_{j\geq 1} s^j = \frac{1+s}{1-s}.$$

Coefficient comparison gives

$$\kappa_{d,\ell} = c_{d,\ell} + 2\sum_{m=0}^{\ell-1} c_{d,m}.$$

Substitution and finite reindexing prove (12). For (10), one has $q_{m+j} = [A(r_m + j) + 1/4]_<$, which proves (13). No floor has been approximated or moved. $\square$

The shifts are

$$r_m \in \begin{cases} \mathbb{N}_0 + \frac{1}{2}, & d \text{ odd}, \\ \mathbb{N}, & d \text{ even}. \end{cases}$$

Thus Conjecture 1.1 is asked only on the two grids that actually occur.

4 The signed Weyl primitive

Define the right-continuous branching staircase

$$S_d(z) = \sum_{\substack{m \geq 0 \\ r_m \leq z}} c_{d,m} \tag{14}$$

and its discrepancy primitive

$$\Delta_d(z) = \int_0^z \left(S_d(u) - \frac{u^p}{p!} \right) \mathrm{d}u. \tag{15}$$

Proposition 4.1 (Signed branching primitive)**.** *For every $d \geq 3$ and $z \geq 0$,*

$$\boxed{\Delta_d(z) \leq 0.} \tag{16}$$

For $d = 3$, equality at positive z occurs precisely at the positive integers. For $d \geq 4$, the inequality is strict for $z > 0$.

Proof. For $0 \leq z < r_0 = p/2$, $S_d(z) = 0$, so

$$\Delta_d(z) = -\frac{z^{p+1}}{(p+1)!} \leq 0.$$

Let $M \geq 0$ and $r_M = M + p/2$. The hockey-stick identity gives

$$\int_0^{r_M} S_d(u)\,\mathrm{d}u = \sum_{m=0}^{M-1} (r_M - r_m) c_{d,m} = \binom{M+p}{p+1}.$$

Therefore

$$(p+1)!\Delta_d(r_M) = \prod_{j=0}^{p} (M+j) - \left(M + \frac{p}{2}\right)^{p+1} \leq 0, \tag{17}$$

because the arithmetic mean of $M, M+1, \ldots, M+p$ is $M + p/2$.

On the half-open cell $[r_M, r_{M+1})$,

$$S_d(z) = \binom{M+p}{p}, \qquad \Delta'_d(z) = \binom{M+p}{p} - \frac{z^p}{p!}.$$

If the critical point lies in this cell, it is

$$z^*_M = \left(\prod_{j=1}^{p} (M+j) \right)^{1/p}.$$

Direct substitution into the cell formula gives

$$(p+1)!\Delta_d(z^*_M) = p(z^*_M)^p \left(z^*_M - M - \frac{p+1}{2} \right) \leq 0, \tag{18}$$

since z_M^* is the geometric mean of $M+1,\ldots,M+p$ and their arithmetic mean is $M+(p+1)/2$. If the critical point is outside the cell, the cell maximum is at an endpoint, already covered by (17). This proves (16).

For $p=1$, the cell maximum is $z=M+1$ and has value zero. For $p\geq 2$, the consecutive factors in both AM–GM applications are unequal, which gives the stated strictness. $\square$

5 Exact defect identity and conditional closure

Define

$$D_A(r) = 2\int_r^K A(z)\,\mathrm{d}z - T_A(r) \tag{19}$$

and the continuous payment

$$W_d = \frac{2}{p!}\int_0^K z^p A(z)\,\mathrm{d}z. \tag{20}$$

Lemma 5.1 (Exact Weyl normalization)**.** *For $p=d-2$,*

$$\int_0^a z^p G_a(z)\,\mathrm{d}z = \frac{\Gamma((p+1)/2)}{4\sqrt{\pi}\,(p+2)\Gamma((p+4)/2)}a^{p+2}. \tag{21}$$

Consequently,

$$\boxed{W_d = \frac{1-\rho^d}{2^d\Gamma(d/2+1)^2}K^d = L_d|A_{\rho,1}^{(d)}|K^d.} \tag{22}$$

Proof. Scale $z=ax$ in (2), integrate the term containing $\arccos x$ by parts, and evaluate the resulting beta integrals. This gives (21). Multiplying the difference of (21) at $a=K$ and $a=\rho K$ by $2/p!$, the duplication formula for Γ reduces the coefficient to

$$\frac{1}{2^{p+2}\Gamma((p+4)/2)^2} = \frac{1}{2^d\Gamma(d/2+1)^2}.$$

The last equality in (22) follows from $|A_{\rho,1}^{(d)}| = \omega_d(1-\rho^d)$. $\square$

Proposition 5.2 (Branching-defect identity)**.** *Let*

$$\mathcal{B}_d(A) = 2\int_0^K (-A'(z))(-\Delta_d(z))\,\mathrm{d}z. \tag{23}$$

Then $\mathcal{B}_d(A)\geq 0$ and

$$\boxed{W_d - P_d^< = \mathcal{B}_d(A) + \sum_{m\geq 0} c_{d,m}D_A(r_m).} \tag{24}$$

Proof. The shell action is absolutely continuous, decreasing, and vanishes at K. Fubini gives

$$\sum_{m\geq 0} c_{d,m}\,2\int_{r_m}^K A(z)\,\mathrm{d}z = 2\int_0^K S_d(z)A(z)\,\mathrm{d}z.$$

Since $\Delta_d' = S_d - z^p/p!$ almost everywhere and both endpoint terms vanish, integration by parts gives

$$2\int_0^K S_dA = W_d + 2\int_0^K A\Delta_d' = W_d - 2\int_0^K A'\Delta_d = W_d - \mathcal{B}_d(A).$$

Both A' and Δ_d are nonpositive, so $\mathcal{B}_d(A)\geq 0$. Subtract (13) and use (19). $\square$

Theorem 5.3 (Conditional all-dimensional shell theorem)**.** *If Conjecture 1.1 holds, then (1) holds for every $d \geq 3$, $0 < \rho < 1$, and $K \geq 0$. It then also holds for every shell $A^{(d)}_{r,R} = \{x \in \mathbb{R}^d : r < |x| < R\}$ and every spectral parameter.*

Proof. The case $K = 0$ is trivial, so assume $K > 0$. Every r_m in (11) lies in $\frac{1}{2}\mathbb{N}$, so the conjecture gives $D_A(r_m) \geq 0$. Equations (9), (24), and (22) yield

$$N_D(A^{(d)}_{\rho,1}, K^2) \leq P^<_d \leq W_d = L_d|A^{(d)}_{\rho,1}|K^d.$$

For a shell with outer radius R, take $\rho = r/R$ and $K = R\sqrt{\Lambda}$ and use Dirichlet dilation. The non-strict counting form follows by applying the strict estimate at $\Lambda + \varepsilon$ and letting $\varepsilon \downarrow 0$. □

6 Proved tail regions and the first interface cell

The action derivative is

$$-A'(z) = \frac{1}{\pi}\begin{cases} \arccos\dfrac{z}{K} - \arccos\dfrac{z}{\mu}, & 0 < z < \mu, \\ \arccos\dfrac{z}{K}, & \mu \leq z < K. \end{cases} \tag{25}$$

Thus A is decreasing and concave on $(0, \mu)$, decreasing and convex on (μ, K), and A, A' match at μ.

Proposition 6.1 (Outer and terminal tails)**.** *Conjecture 1.1 holds in either of the following cases:*

1. *$r \geq \mu$;*
2. *$G_K(r) \leq 3/4$.*

Proof. If $r \geq \mu$, then $A = G_K$ on $[r, K]$; the claim is trivial when $r \geq K$. Otherwise set $B = \lceil K - r\rceil$ and zero-extend

$$g(s) = G_K(r + s) \quad (0 \leq s \leq K - r)$$

to the integer interval $[0, B]$. Since $G_K(K) = G'_K(K) = 0$, this extension is nonnegative, decreasing, convex, 1/2-Lipschitz, and vanishes at B. The convex shifted-floor theorem of [1] therefore bounds its ordinary-floor tail by $2\int_0^B g$. At every sample,

$$[g(j) + 1/4]_< \leq \lfloor g(j) + 1/4 \rfloor.$$

The strict tail is consequently bounded by $2\int_0^B g = 2\int_r^K G_K$.

If $G_K(r) \leq 3/4$, monotonicity and $A \leq G_K$ imply $A(r + j) + 1/4 \leq 1$ for every $j \geq 0$. Every strict bracket is therefore zero. Notice that equality at 1 is harmless precisely because the bracket is strict. □

We next sharpen the prescribed first milestone. For the ball action set

$$T_K(r) = [G_K(r) + \tfrac{1}{4}]_< + 2\sum_{j\geq 1}[G_K(r + j) + \tfrac{1}{4}]_<, \qquad D_K(r) = 2\int_r^K G_K(z)\,\mathrm{d}z - T_K(r). \tag{26}$$

The wall-safe translation in the preceding proof gives $D_K(r) \geq 0$.

Lemma 6.2 (A low-action 1/3-Lipschitz reserve)**.** *Let g be strictly decreasing and convex on $[0,b]$, with $g(b)=0$ and $\mathrm{Lip}(g)\le 1/3$. If*

$$\frac{3}{4} < g(0) < 1,$$

then

$$2\int_0^b g(z)\,\mathrm{d}z - \left([g(0)+\tfrac14]_< + 2\sum_{j\ge1}[g(j)+\tfrac14]_<\right) > \frac{11}{16}. \tag{27}$$

The function is understood to be zero-extended after b.

Proof. Let $h=g(0)$ and let $x\in(0,b)$ be defined by $g(x)=3/4$. Since $g(j)<1$, a sample contributes exactly one if and only if $j<x$. Therefore the bracketed count is

$$2\lceil x\rceil - 1.$$

The Lipschitz bound and $g(b)=0$ give

$$g(z)\ge\max\{h-z/3,0\},\qquad 2\int_0^b g(z)\,\mathrm{d}z \ge 3h^2 > \frac{27}{16}.$$

If $x\le1$, the count is one and (27) follows.

Suppose $x>1$ and put $a=(h-3/4)/x$. Convexity implies that, for $z\ge x$,

$$g(z)\ge\max\left\{\frac34 - a(z-x), 0\right\}.$$

Hence

$$\int_0^b g(z)\,\mathrm{d}z \ge \frac34 x + \frac{(3/4)^2}{2a} = x\left(\frac34+\frac{9}{32(h-3/4)}\right) > \frac{15}{8}x.$$

Since $2\lceil x\rceil-1<2x+1$, the defect is larger than

$$\frac{15}{4}x-(2x+1)=\frac74 x-1>\frac34>\frac{11}{16}.$$

□

Proposition 6.3 (Exact one-interface reduction)**.** *Assume*

$$0<\mu-r<1,\qquad r\in\tfrac12\mathbb{N}.$$

Define

$$n_0=[G_K(r)+\tfrac14]_< - [G_K(r)-G_\mu(r)+\tfrac14]_< \in\mathbb{N}_0 \tag{28}$$

and

$$\delta_\mu(r)=\int_r^\mu G_\mu(z)\,\mathrm{d}z.$$

Then

$$\boxed{D_A(r)=D_K(r)+n_0-2\delta_\mu(r).} \tag{29}$$

Moreover,

$$2\delta_\mu(r) < \frac{4(\mu-r)^{5/2}}{15\sqrt{\mu}} < \frac{4\sqrt2}{15} < 1. \tag{30}$$

Consequently, the shifted-tail inequality holds if any of the following is true:

1. $n_0 \geq 1$*;*

2. $G_K(\max\{r, K/2\}) \geq 1$*;*

3. $r \geq K/2$*.*

The only remaining one-interface configurations satisfy

$$\boxed{r < \frac{K}{2}, \qquad K < \frac{1}{\omega_0}, \qquad n_0 = 0, \qquad \omega_0 := G_1(1/2) = \frac{\sqrt{3}}{2\pi} - \frac{1}{6}.} \tag{31}$$

In particular,

$$r \in \left\{\frac{1}{2}, 1, \frac{3}{2}, 2, \frac{5}{2}, 3, \frac{7}{2}, 4, \frac{9}{2}\right\}.$$

Proof. Since $r + j > \mu$ for every $j \geq 1$, all samples after the first agree with the ball samples. Thus $T_A(r) = T_K(r) - n_0$. Subtracting $2\int_r^\mu G_\mu$ from the ball integral proves (29).

The turning estimate

$$G_\mu(z) < \frac{(\mu - z)^{3/2}}{3\sqrt{\mu}} \qquad (0 < z < \mu)$$

integrates to (30), using $\mu > r \geq 1/2$ and $\mu - r < 1$. If $n_0 \geq 1$, then $D_K(r) \geq 0$ and (29) is positive.

For $r < K$, use the same integer endpoint $B = \lceil K - r \rceil$ and zero-extended function $g(s) = G_K(r + s)$. Put

$$s_* = \max\{0, K/2 - r\}.$$

Then g is 1/3-Lipschitz on $[s_*, B]$ and $g(s_*) = G_K(\max\{r, K/2\})$. The refined ordinary-floor theorem of [3, Theorem 3.4], followed by $[\,\cdot\,]_< \leq \lfloor \cdot \rfloor$, gives

$$D_K(r) \geq \frac{1}{2} \lfloor G_K(\max\{r, K/2\}) \rfloor. \tag{32}$$

For $r \geq K$ the same inequality is trivial. Hence the second condition and (30) prove the claim.

If $r \geq K/2$ and $G_K(r) < 1$, the translated ball action is globally 1/3-Lipschitz. Its tail is either zero or Lemma 6.2 applies. In the latter case $D_K(r) > 11/16 > 4\sqrt{2}/15$, so (29) is again positive. This proves the third condition.

If none of the three conditions holds, then $r < K/2$ and $G_K(K/2) = \omega_0 K < 1$, giving (31). The elementary rational bounds

$$\frac{265}{153} < \sqrt{3} < \frac{26}{15}, \qquad \frac{333}{106} < \pi < \frac{355}{113}$$

(the square-root bounds follow by squaring) give $1/10 < \omega_0 < 1/9$. Consequently

$$\frac{9}{2} < \frac{1}{2\omega_0} < 5.$$

Since $r < K/2 < 1/(2\omega_0)$ and $r \in \frac{1}{2}\mathbb{N}$, only the nine displayed shifts remain. □

Remark 6.4 (What remains in the first milestone)*.* Proposition 6.3 does not finish the one-interface theorem. It replaces a two-parameter fractional-interface problem by a bounded nine-shift residual family: $2r < K < 1/\omega_0$, and $r < \mu < \min\{r + 1, K\}$, with the explicit no-loss condition $n_0 = 0$. The required inequality there is exactly

$$D_K(r) \geq 2\delta_\mu(r).$$

This is the next scalar reserve lemma to prove.

7 Cell recurrence and the corrected global target

For $x_j = r + j$, set

$$Q_j = [A(x_j) + \frac{1}{4}]_<, \qquad D_j = 2\int_{x_j}^{K} A(z)\,\mathrm{d}z - \left(Q_j + 2\sum_{k>j} Q_k\right).$$

Then

$$D_j - D_{j+1} = 2\int_{x_j}^{x_j+1} A(z)\,\mathrm{d}z - Q_j - Q_{j+1}. \tag{33}$$

If

$$e_j = A(x_j) + \frac{1}{4} - Q_j \in (0, 1]$$

and

$$C_j = 2\int_{x_j}^{x_j+1} A(z)\,\mathrm{d}z - A(x_j) - A(x_j + 1),$$

then

$$D_j - D_{j+1} = C_j - \frac{1}{2} + e_j + e_{j+1}. \tag{34}$$

On a fully inner cell,

$$C_j = -\int_0^1 u(1-u)A''(x_j + u)\,\mathrm{d}u \geq 0.$$

Individual increments in (34) can nevertheless be negative, and a negative increment need not be paid by the next floor-drop cell. Thus the viable block formulation must retain the terminal convex reserve. If J is the first index with $x_J \geq \mu$, the exact target is

$$\boxed{D_0 = \sum_{j=0}^{J-1}(D_j - D_{j+1}) + D_J \geq 0,} \tag{35}$$

where $D_J \geq 0$ is the translated convex ball-tail defect. A proof of (35) needs a quantitative, not merely nonnegative, lower bound for D_J or an equivalent matched shelf reserve.

8 A dimension-dependent no-mode region

The radial form (6), the one-dimensional Dirichlet Poincaré inequality, and $r^{-2} \geq 1$ on $(\rho, 1)$ give

$$\lambda_{\ell,n} \geq \frac{n^2\pi^2}{(1-\rho)^2} + \nu_{\ell,d}^2 - \frac{1}{4}.$$

Since $\nu_{0,d} = (d-2)/2$, one obtains the strengthened wall

$$N_D(A_{\rho,1}^{(d)}, K^2) = 0 \quad \text{if} \quad K^2 \leq \frac{\pi^2}{(1-\rho)^2} + \frac{(d-2)^2 - 1}{4}. \tag{36}$$

The equality face belongs to the no-mode region because the counting function is strict.

9 Remaining proof obligations

The primary route now consists of the following precise steps.

1. Prove the bounded nine-shift residual inequality following Proposition 6.3; this completes the first interface-cell milestone.

2. For arbitrary $r < \mu$, prove the shelf-block inequality (35), retaining a quantitative convex suffix reserve. A one-negative-cell/one-drop pairing is not a valid target.

3. Insert the resulting positive shifted-tail theorem into Theorem 5.3. No dimension-dependent angular or optical theorem is then needed.

If the second step fails only for an explicitly controlled family of tails, the exact identity (24) is the appropriate aggregate fallback: the weighted negative part must be compared directly with $\mathcal{B}_d(A)$.

References

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